

Final Project: IITB Insti-Assist — A RAG-Powered AI Assistant for IIT Bombay

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Final Project Write-Up — Academic Assistant

1 Chosen Scope and Why

This project implements the **Academic Assistant** scope: A RAG-powered AI assistant that answers questions about IIT Bombay's academic policies - grading, registration, exam rules, malpractice rules, degree requirements etc. by retrieving relevant sections from official institute rulebooks and generating an LLM's answer by using them as context. If the answer isn't supported by the retrieved documents, the assistant says "I don't know" instead of guessing.

This scope was chosen because I feel it's the most basic thing a student is worried about when coming to a college - what all does the institution holds in its academic parts. I felt this could be quite helpful, and also it was comparatively easier to get documents relevant to this!

2 Data Sources Used

Ten official IIT Bombay PDF rulebooks were used as the knowledge base, covering rules across different degrees and also information about academic conduct:

- `ugrulebook.pdf` - Undergraduate academic rules
- `Mtech_MBA_rules.pdf` - M.Tech / MBA academic rules
- `phd_rules.pdf` - PhD academic rules
- `M_Eng_Masters_Rules.pdf` - M.Eng / Masters rules
- `IDDDP_rules.pdf` - Inter-Disciplinary Dual Degree Programme rules
- `grading.pdf` - Grading policy
- `academic_calendar.pdf` - Academic calendar
- `award_rules.pdf` - Award rules
- `acad_malpractices_1.pdf` - Academic malpractice policy (part 1)
- `acad_malpractices_2.pdf` - Academic malpractice policy (part 2)

These aren't the original names of the files on the web as the files were renamed, but these are the official rulebooks easily accessible through internet.

3 Chunking Strategy and Why

All the documents were split into fixed-size chunks: 200 characters per chunk with a 40-character overlap between consecutive chunks. Each chunk retains a reference to its source PDF filename, so that the source is easily printable with the final answer. This approach

was chosen for its simplicity and predictability - it requires no sentence-boundary detection or additional NLP preprocessing, and prevents information loss at chunk boundaries.

There was a slight trade-off that 200 characters is a fairly short window (roughly 30-40 words), so chunks can occasionally cut a sentence or clause mid-way, which slightly reduces the semantic coherence of any single chunk, but was still adopted by me as a reasonable trade-off for a one-week project, given the overlap partially compensates for it.

4 Known Limitations and Future Improvements

- **Chunk coherence:** A 200 token chunk means character-based chunking can still split sentences mid-way, leading to some lack in semantic coherence. A future version could use sentence or paragraph-aware chunking to preserve semantic units.
- **No conversational memory:** There is no memory of the previous chats already taken with the assistant, i.e. each question has to be new and fully framed independently and can't be an add on to a previously asked question
- **Threshold tuning:** The 0.25 similarity threshold was set from a small manual test set of on-topic and off-topic questions. It may not be the best choice among different values, some relevant sources may be cut-off or some irrelevant sources may get included.
- **No citation-level highlighting:** Sources are shown at the file level, not as exact quoted snippets, so a user cannot immediately see which specific passage supported an answer. Though this isn't a very difficult addition in my current code itself, due to lack of time I refrained from doing it on the first go.
- **No confidence-score printed:** The answer is not shown with a confidence score, that is how much the assistant is sure about it being near to actual thing. Again this is an easy addition in my code structure, but due to time constraints wasn't added by me as of now!
- **Lack of complete information:** The pdfS may not be exhaustive of institute's policy as only 10 of them are used which seemed to be fairly ok for a week's project, but much more additions can make the assistant even better informed.

5 Sample-Output Image

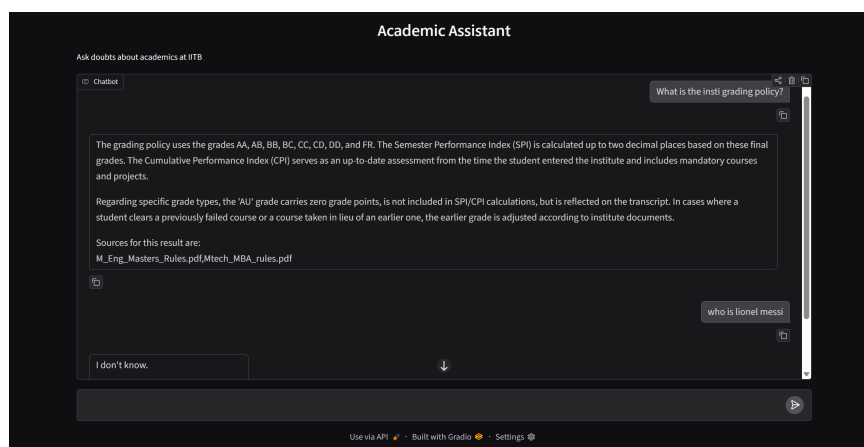


Figure 1: One of the Assistant's output during testing